WRRA Core 2.0

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13 October 2026

WRRA aims to explain one nonprivileged universe through a single method based on minimum computation, the common carrier, and phenotype. Core 2.0 retains this research aim and specifies an execution framework that can also be explicitly mapped to other domains. Numerical values and physical completion assumptions calibrated to our universe are recorded in WRRA_M.

Core 1.0 defined a common grammar of present state, residue, boundary, common carrier, update, renderer, phenotype, and record. Core 2.0 inherits that grammar and specifies interfaces for preparation, classification and allocation, correlation preservation, resource authorization, frame readout, and residue storage. Its central change is to turn an explanatory sequence into an auditable execution procedure.

Definition in one sentence

Starting from finite present inputs and resources, classify and allocate a prepared state, perform only the necessary updates through a common carrier that preserves ownership and correlations, let a resource-authorized renderer produce phenotypes and records, and preserve unprocessed state and remaining resources as residue in the present.

Reading order

Sections 1–2 describe inherited principles and execution types; Sections 3–7 specify the new interfaces; Section 8 defines assessment rules. Sections 9–10 give numerical evidence and boundaries for the current physical implementation. Sections 11–12 address domain mapping, version management, conformance, and provenance.

Completion of the Core specification is distinct from completion of the physics. What is specified as version 2.0 is the common execution grammar. This does not establish the uniqueness of every physical assumption, selection of a single quantum outcome, primordial cosmological perturbations, or a restart mechanism.

Version declaration

As of 13 October 2026, WRRA Core 2.0 inherits minimum computation, the common carrier, phenotype, the present, ownership, boundaries, and ledgers, and specifies preparation, classification, allocation, resource authorization, frame readout, and residue storage as required execution contracts. Core 1.0 is preserved as an earlier specification; this document is a separate successor.

1 Inherited principles and version changes

Minimum computation

Select the minimally sufficient structure that satisfies the declared contracts for expression, survival, accuracy, and stability. Required memory, reserves, and redundancy belong to the cost of maintaining those contracts. A small implementation alone does not establish minimality; compare costs, errors, and functions against simpler alternatives.

$$A^* \in \mathrm{ParetoMin}\ \{\ C(A) : A \text{ satisfies the declared execution contract}\ \}$$

Common carrier and phenotype

Multiple channels share a reusable transport and update structure while retaining channel ownership and differences in readout. Internal state becomes phenotype through a renderer. Source, possible state, load, energy, and observable are not interchangeable variables without an explicit transformation.

The present and finiteness

Past effects operate through state, residue, and accessible records that remain in the present. Addresses, active distinctions, resources, and record capacity used in execution must be finite and specified for the implementation. Infinite resources are not admitted as available execution resources; unallocated resources are Null. This rule does not imply that a particular address count is a unique constant of nature.

Item	Core 1.0	Core 2.0 specification
Before execution	SOURCE and STATE	Preparation contract and input provenance
Transport	Reuse of a common carrier	Distinct classification, allocation, channel, and mode indices
Residue and record	Effects remaining in the present	Storage contract for state, pending input, and budget
Update	UPDATE and RESET	Propose, check budget, authorize, commit, or stall
Renderer	Internal state to phenotype	Distinguish readout, recording, and single-outcome selection
Assessment	Separate interpretation, computation, and prediction	Explicit known-value reproduction and unmeasured outputs

The frozen Core 1.0 remains an earlier reference. Version 2.0 extends required execution grammar while retaining the constitutional invariants. Changing a physical coefficient alone does not advance the Core version. Making preparation, authorization, and storage interfaces mandatory is the basis for the major version change. [1, 2]

2 Execution state and index types

Core state

The following expression is common notation introduced in 2.0 to connect existing models. It must not be presented as a newly derived microscopic law for a particular domain.

$$Z = (S, \Pi, X, R, B, C, Q, O, L)$$

S is the present input and source; Π is the preparation contract; X is internal state; R is residue; B is the boundary; C is the common carrier; Q specifies resources and capacity; O is the readout rule; and L is the provenance, conservation, and outcome ledger. The update index k denotes execution order. A domain must specify a clock and units before treating it as physical time.

Element	Required execution specification
SOURCE	Inputs, boundary inputs, provenance, and active range
PREPARATION	Distribution, basis, conditional state, and resource preparation
RELATION LAW	Allowed transformations, couplings, constraints, and classification rules
STATE RESIDUE	Joint state, correlations, and present effects
BOUNDARY	Internal and external owners, inflow, outflow, and allowed routes
COMMON CARRIER	Shared structure, channel owners, and transport operator
RESOURCE GATE	Cost, capacity, authorization, and shortage handling
UPDATE RENDERER	State transition and internal-to-phenotype transformation
PHENOTYPE OBSERVABLE	Actual output and measurable readout
RECORD LEDGER STORE	Records, balances, and storage locations and references

Distinct indices

Address n, allocation route j, carrier mode k, channel c, physical coordinate x, and observation outcome y are different types. Equal cardinalities do not make two types identical. A mapping between types must declare its transformation, units, conserved quantities, and scope.

In the current example, 1,015,000 is an address range; 128 is an allocation-slot count or the dimension of a separately connected mode register. A mod15 group is an arithmetic classification, not verification of a CCS-15 physical channel. The 128×2×2 register comprises 512 joint source, supply, and record states in one specific construction. [2–7]

3 Prephenotypic classification and allocation

Validation inputs

Declare a finite address or item set, nonnegative weights, classification effects, preparation states, and route capacities. Distinguish verified values, construction assumptions, and diagnostic inputs. Static prephenotypic load does not automatically carry a rate, physical-space coordinate, or energy unit.

WRRA transformation

A classifier records admitted, excluded, and pending items. Allocation may follow actual model load rather than address counts. The domain must specify whether a contribution can be divided and whether route capacities are equal.

$$c_s(n) = w_n\, e_s(n)\, g_s(n), \quad m_n = \Sigma_s\, c_s(n)$$

$$T_{jn} \geq 0, \quad \Sigma_j\, T_{jn} = 1, \quad M_j = \Sigma_n\, T_{jn}\, m_n$$

Excluded items remain in an exclusion ledger outside the allocation target. A claim of preserving all inputs requires complete accounting across active, excluded, and pending items. g_s is a response weight; using it as a probability requires a separate normalization contract.

Arithmetic sieve reference implementation

$$H_p(n) = 1 - (1/p)\, \Sigma_{h=0\ldots p-1}\, \exp(2\pi i h n/p)$$

$$r_P(n) = 1_{\text{composite}}(n)\, \Pi_{p\in P}\, H_p(n)$$

Composite candidates remaining in a partial sieve, removal of composites by a complete classifier, and failure of normal return in a two-stage Actual process are different residues. They must not be added under one label. The Core does not require every domain to use a prime sieve.

Outputs

An allocation that keeps each address indivisible has an imbalance lower bound determined by the largest atomic load. The divisible-contribution construction v0.14 equalized total load across 128 routes while retaining correlations. The sieve-plus-atomic-allocation construction v0.1 reduced concentration without achieving perfect equality. These are outcomes of different divisibility contracts.

Falsification conditions and assessment

Negative allocation, omissions or duplicates, altered totals, deletion of the exclusion ledger, or loss of correlations falsifies the interface. If equal load is imposed by the capacity rule, equality is assessed as a construction result. It is not promoted to a derivation of small cosmological perturbations. [3–5]

4 Correlation preserving state and response

Validation inputs

Read address weights w_n, sector effects e_s and responses g_s, address-conditioned carrier states ρ_n, and readout operators A_s from the same ledger. Marginal distributions that retain only totals generally cannot reconstruct address–state correlations.

WRRA transformation

$$f_s(n) = e_s(n)\, g_s(n)$$

$$\Sigma_s = \Sigma_n w_n f_s(n) \rho_n, \quad L_s = \mathrm{Tr}(\Sigma_s A_s)$$

Σ_s is an unnormalized positive weighted operator. A load whose sum differs from one is not a normalized probability. When grouping addresses into C_a to reduce calculation, retain conditional states τ_a.

$$B_{saj} = \Sigma_{(n \in C_a)} T_{jn} w_n f_s(n)$$

$$\Sigma_s = \Sigma_{ja} B_{saj} \tau_a$$

If address states differ within a group, verify that averaging preserves the required readout exactly or refine the group. Reducing address count alone does not justify claiming equivalence.

Decomposition of frame outcomes

Store outcome-conditioned means and covariances. Recover the total covariance from within-branch and between-branch terms.

$$\mathrm{Cov}(b) = \Sigma_y p_y \mathrm{Cov}(b|y) + \mathrm{Cov}_y(E[b|y])$$

Averaging only within-branch covariance loses correlations associated with differences in branch means. Nonselective disposal of a record does not always destroy moments; identify the actual discarded information in the reduction.

Response ownership and contracts

When defining response through second derivatives of a preparation generating function Ψ, distinguish joint fluctuation and address-clamped contracts. Classical or commuting cases may use the corresponding covariance expression; noncommuting cases require an appropriate response definition. An internal Hessian does not become gravitational stress or an effective coupling constant without a deformation and unit mapping.

Falsification conditions and assessment

Disagreement between direct address sums and grouped calculations, loss of positivity, omission of between-branch covariance, or mixing responses with different owners falsifies the construction. Correlation preservation is a computational Core-conformance result; physical-space generation requires separate evidence. [3, 4, 6]

5 Renderer shutter and record

Validation inputs

Specify the prepared state, readout operator or measurement instrument, outcome-record factors, accessible timing information, and exposure window. A frame is a specified readout state; frame interval and exposure duration are not automatically identical. Prephenotypic allocation order need not be assigned physical time.

WRRA transformation

The renderer supplies a map from internal state to actual output and record. A quantum domain may specify completely positive outcome instruments I_y with $p_y = \mathrm{Tr}\, I_y(\rho)$ and $\Sigma_y p_y = 1$. This is a domain choice; quantum instruments are not imposed on other fields.

A time window averaged under the same Hamiltonian provides the following example.

$$[S_\tau(\rho)]_{ij} = \rho_{ij}\, \mathrm{sinc}[(E_i - E_j)\tau/(2\hbar)]$$

This is conditional readout for a centered uniform window. $\mathrm{sinc}(z)=\sin(z)/z$, with value one at $z=0$. Diagonal energy occupations and mean energy under the same Hamiltonian remain unchanged. Retaining or discarding timing labels may change coherence readout.

Three distinct results

First, one may calculate which distinctions a window average blurs. Second, one may calculate outcome probabilities and conditional records. Third, one may provide a rule that selects an actual single outcome. The first two results do not automatically establish the third.

Outputs

Construction v0.22 connected the inherited 128-mode SI calibration to the shutter window. Diagonal preparations remained unchanged, whereas phase-sensitive readout of coherent controls changed. Construction v0.23 calculated the influence of access to timing labels on readout blurring. The preparation-metadata selector in v0.25 exposed finite probability-resolution and linearity limits.

Falsification conditions and assessment

Incorrect occupation or energy changes under same-H averaging, or disagreement between direct time averaging and the sinc expression, falsifies the calculation. A preparation-metadata policy must not be presented as a universal linear quantum channel. The shutter result explains the structure of readout and accessible information; it does not complete a derivation of universal uncertainty, the Born rule, or removal of all ultraviolet divergences. [6–8]

6 Resource authorization and frame exchange

Validation inputs

Declare source and supplier states, cost functions, record capacity, the full boundary, and a Hamiltonian or domain-specific resource-accounting rule. A transition that increases source readout requires an owner for the supplied increment. Distinguish energy-degenerate logical records from the cost of an actual measurement pointer.

WRRA transformation

Execution proceeds through transition proposal, cost calculation, budget and capacity checks, and commitment of an authorized transition. An unspecified supplier gives unresolved_supplier; insufficient resources give stall. Neither condition allows that proposal to overwrite the state. Concurrent committed transitions, if present, are recorded separately.

For a specific quantum implementation, conservation may be specified as:

$$\Gamma_{after} = U \Gamma_{before} U^\dagger$$

$$H_{total} = H_{source} + H_{supply} + H_{record} + H_{interaction}$$

$$[U, H_{total}(V)] = 0$$

If commutation holds for every declared V, test energy preservation and preservation of the volume derivative of that same Hamiltonian. A fixed-volume test alone does not close all expansion and interaction boundaries.

Current finite exchange example

$$|8, B1, R0\rangle \leftrightarrow |72, B0, R1\rangle$$

Apply this exchange in a 512-state register comprising a 128-mode source, a supply bit, and a logical record bit; leave all other basis states unchanged. The supply gap provides the energy difference between modes 8 and 72. This is a construction with specified supplier preparation and coupling.

Outputs

Account for increased source readout and decreased supply energy in one ledger. Readout can change without creating additional total energy. Full logical observables transported as $O_{after}=U O_{before} U^\dagger$ may remain preserved even though fixed source-only readout A changes.

Falsification conditions and assessment

Omitted or double-counted suppliers, commitment without authorization, concealed record overflow, pressure inconsistent with the same energy function, or loss of full observables falsifies the construction. A gravitationally active supplier belongs in the full physical stress source. Resource closure does not replace a law generating the natural supplier. [2, 9]

7 Residue storage and the next present

Validation inputs

Receive the post-transition joint state, outcome records, unprocessed inputs, remaining supply budget, conditional correlations, and storage references. Retain the Core 1.0 principle that earlier events are accessed as presently owned records.

WRRA transformation

The storage contract preserves the following items together. Distinguish copying physical states from checkpointing calculated state descriptions.

Stored item	Required preserved content
Present state	Source, supply, record, and joint correlations
Residue	Ownership of failed normal return and remaining effects
Pending input	Unprocessed items and their state
Budget	Remaining energy, resources, and record capacity
Readout reference	Operators, bases, units, and preparation contracts
Provenance and version	Input files, rules, hashes, and execution results
Execution status	Distinction between commit, stall, and unresolved

The common update notation is:

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(Z_(k+1), outcome, ledger) = F_D(Z_k, input, declared_contract)
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The domain specifies the actual law F_D. Interpreting order index k as a physical clock requires a separate mapping. Residue storage preserves information for subsequent execution; it does not generate new resources or blank records.

Restart interface

The current FrameResidualStore uses restart_condition = black_box_unspecified and restart_transition = null. Core 2.0 supplies no restart threshold, time, period, or next-Big-Bang condition for this implementation. A domain that actually executes reset must separately specify its trigger, cost, transition, and record preservation.

Outputs and falsification conditions

Preservation of actual correlations, records, pending inputs, and remaining resources closes the storage contract. Automatically treating numerical error as physical residue, restoring depleted supply during storage, or invoking an unspecified restart falsifies the contract. The completed scope is storage and reference preservation; cosmic reactivation remains open. [1, 2, 9]

8 Assessment rules and prediction ownership

Every assessment first states validation inputs → WRRA transformation → outputs → falsification conditions, and then gives its conclusion. The explanatory document and computation ledger must refer to the same inputs, rules, and outputs.

Inputs and outputs

Revising a model using verified constants and observations is legitimate theory construction. Distinguish verified values, calibration values, construction assumptions, and diagnostic conditions. When fitting one quantity and calculating another, identify both the fitted input and the resulting output.

Contract	Result to assess
Structural explanation	WRRA-specific connections, transformations, and owners
Computation	Reproducible output from declared inputs and rules
Known-value reproduction	Consistency with reality and explanatory achievement
Unmeasured prediction	An unmeasured quantity calculated by the fixed model
Empirical validation	An appropriate comparison with observations or experiments
Control	A declared action changes goals or survival conditions
OPEN PARTIAL	Required inputs, transformations, or checks remain incomplete

Originality and prediction

Independent predictions are not required for WRRA to qualify as a theory or to establish originality. Assess originality through WRRA-specific structure, transformations, connections, and explanation rather than numerical novelty. Prior knowledge of a value does not invalidate its reproduction.

After fixing a model using verified values, an unmeasured physical quantity actually calculated by that model is assessed as a WRRA prediction. If another theory yields the same number, WRRA retains ownership of its prediction when it calculated it through its own structure. Merely copying an external result without calculation is different from an internal output.

External theories and evidential standards

External theories, observations, and numerical tools supply validation and reinforcement. An element is integrated only to the extent that it actually executes in the final WRRA calculation path. Completion of a microscopic structural derivation is a separate claim. Apply the same disclosure, reproducibility, error, and falsification standards to WRRA and existing theories.

Scope of failure

Preserve a failure at the level of the affected connection, candidate, or implementation. A failed control does not automatically reject WRRA as a whole; a passed conservation check does not automatically validate all observations.

9 Current physical calculation path

Validation inputs

The reference implementation is WRRA Finite Frame Bridge v0.29.1. It uses N=1,015,000 addresses, inactive address 1, 128 modes, weighted and unweighted branches, three sector effects, and finite supply and record registers. The address count is a frozen construction input for reproduction, not an observationally established unique constant.

WRRA transformation

Arithmetic effects and conditional states → correlation-weighted operators → finite source–supply exchange → load, density, and pressure from the same energy function → the total background and conditional physical readout.

$$E_s = \eta_s V_0 a^{(\nu_s)} L_s, \quad u_s = E_s/(V_0 a^3)$$

$$p_s = -\partial E_s/\partial V \,|_L = -(\nu_s/3)\, u_s$$

The current ν=(0,0,3) is a physical construction input. Distinguish the partial derivative at fixed load from a total derivative that includes state changes. η specifies calibrated SI conversion coefficients, not universal Core constants.

Outputs

Preparation and branch	H_0 km s^{-1} Mpc^{-1}	First three TT peaks ℓ
Uniform supply weighted	67.44991813	220 536 812
Uniform supply unweighted	66.90537863	221 539 818
Gibbs supply weighted	56.47774119	First peak 236
Gibbs supply unweighted	56.02363487	First peak 237

At a=1 and V_0=1 m^3, the transferred energy for weighted uniform preparation is $4.967352081978407\times10^{-10}$ J. It already exists in the prepared supply ledger; exchange does not add new total energy.

Falsification conditions and assessment

Failure of commutation, total energy conservation, volume-derivative consistency, or full information moments under the same inputs falsifies this construction. Known-value reproduction under uniform preparation is a consistency and explanatory result. Different Gibbs-control outputs show that arithmetic alone does not force the preparation state.

P denotes phenotype, D nonphenotypic residue, and R return. Classification probabilities and SI energy fractions are distinct. Absence of a phenotype record does not remove D or R from the total source ledger.

The current conditional CMB calculation actually uses CAMB 2.0.4, standard GR, flatness, adiabatic initial perturbations, and recombination. It is neither the completed CMB of the new interacting mode nor a purely arithmetic derivation of the primordial spectrum. [2, 9]

10 Physical checks and open boundaries

Validation inputs and WRRA transformation

The single-scalar test specifies k=(0.05,0,0) Mpc^{-1}, amplitude 10^{-8}, a=0.02…1, spatial preparation, standard GR, and an explicit transfer recipient. Vacuum-like $T_R^{(\mu\nu)}=-u_R g^{(\mu\nu)}$ requires $Q_R^{\nu}=-\nabla^{\nu}u_R$. The completion selects $Q_D=-Q_R$ and $Q_b=0$. Required compensating transfer and the choice of D as its recipient are separate statements.

Outputs and falsification conditions

Check	Weighted	Unweighted
00 constraint residual with transfer	9.9924×10^{-9}	1.03691×10^{-8}
Transfer omitted under the same normalization	0.999992	0.999992
Spatial trace residual	3.49×10^{-16}	5.77×10^{-16}

The 00 normalization is $\max|F|/\max|S_R|$, using the same vacuum-perturbation source denominator for the positive and omitted-transfer controls. These are finite numerical residuals, not observational sigma or likelihood values. Conservation failure, nonconvergence, or failure of amplitude linearity falsifies this conditional bridge.

Reexecution on 13 October 2026

The 26/26 checks in single_mode.py and 28/28 checks in review.py passed again during preparation of the Korean specification. The earlier 99 frame and background checks and 24 storage checks are cited as released-ledger evidence. This does not report a new execution of the complete CAMB baseline.

Open items

A physical three-dimensional preparation law, primordial amplitude and spectrum, interaction action, directional deformation and anisotropic stress, single quantum-outcome selection, natural supply preparation, and a restart law remain incomplete. The Core identifies the required contracts without assigning arbitrary values.

Observational comparison scope

Official Planck $-2\ln L$ and a BAO covariance likelihood have not been calculated. $D/H=2.44177424\times10^{-5}$ and helium mass fraction 0.245667206 are inherited PRIMAT interpolation values, not outputs of a new WRRA nuclear reaction network. Flatness $\Omega_{total}=1$ is an input condition. Missing observational contracts remain OPEN. [2, 9]

11 Domain mapping and version management

Core and the model of our universe

Core 2.0 is the execution specification; WRRA_M is the numerical and physical profile specialized to our universe; MCC 3.0 is the integrated cosmological manuscript layer. Advancing the Core number does not automatically revise existing WRRA_M materials or revalidate every connection.

Profile	Elements to map in Core 2.0
Physics WRRA_M	Preparation, common carrier, energy, stress, measurement, and record
Bio Cell	Molecular source, executor, membrane boundary, resources, and lineage residue
AI Worm Chess	Sensation, internal state, action, budget, failure, and memory
Econ	Transactions, credit, shocks, debt, resources, and present residue
Met	Atmospheric state, transport, boundaries, readout, and history compression
Math Method	Finite inputs, transformations, proof scope, errors, and records

These are mapping interfaces, not a claim that every domain is implemented and validated under 2.0. Existing Core 1.0 applications retain their profile-specific assessments. Chess is identified as an application target; no new performance validation is added here.

Ownership of physical reference values

CCS-15 is an important physical common-carrier reference structure. Fifteen channels are not mandatory for all domains. The 128 slots, 1,015,000 addresses, 5/26.8/68.2 proportions, shutter times, electron calibration, and neutrino input belong to physical profiles. The speed of light c is a fixed verified input and must not be presented as a prediction.

Migration rules

Retain profile-specific owners and equations while adding preparation contracts, index types, divisibility assumptions, resource authorization, storage, and assessment ledgers. For an inapplicable element, state the reason and the equivalent contract. Do not retroactively label earlier calculations as Core 2.0-validated when they did not use the new specification.

Subsequent revisions

Changes in inputs, calibration, or datasets revise a profile. Wording, notation, and provenance corrections are patches to 2.0. Changes in required execution interfaces receive a successor Core version. If a restart or single-outcome law is implemented, first document its profile-level calculation and falsification conditions, then decide whether the common specification needs extension. [1, 2]

12 Conformance criteria and provenance

Core 2.0 conformance

An implementation must disclose these contracts. Conformance means compliance with the execution specification, distinct from empirical approval of a domain law.

ID	Required contract
C01	Distinguish provenance, calibration, assumptions, and diagnostic inputs
C02	Specify finite active range, preparation, and present-residue owners
C03	Distinguish address, route, mode, channel, coordinate, and outcome types
C04	Preserve classification and allocation totals and exclusion and pending ledgers
C05	Preserve correlations or disclose reduction errors and conditions
C06	Identify shared common-carrier structure and ownership
C07	Specify transition cost, supplier, capacity, authorization, and stall
C08	Distinguish renderer, phenotype, measurement, record, and storage
C09	Include balances from the same resource function and boundary flows
C10	Disclose minimality comparisons and accuracy and function contracts
C11	State inputs → transformation → outputs → falsification → assessment
C12	Preserve executable files, versions, inputs, errors, results, and hashes

Provenance ledger

[1] WRRA_Core_1.0_동결본_응용계보포함_최원식_2026-09-07.docx. Constitutional invariants, required elements, domain separation, and application lineage.

[2] WRRA_Finite_Frame_Bridge_v0_29_1.zip. Manuscript_KO.md, single_mode.py, review.py, and result ledgers.

[3] WRRA_Original_Sieve_Residue_and_Correlated_Interface_v0_12.zip.

[4] WRRA_Prephenotype_Static_Slot_Load_Routing_v0_14.zip.

[5] WRRA_Sieve_Adaptive_Slots_Prephenotypic_v0_1.zip.

[6] WRRA_Frame_Record_P6_Moment_Audit_v0_19.zip.

[7] WRRA_Inherited_128_Mode_Spectrum_Shutter_v0_22.zip.

[8] WRRA_Shutter_Timing_Information_Comparison_v0_23.zip and WRRA_Preparation_Residue_Correlation_v0_25.zip.

[9] WRRA_Fixed_Point_and_Energy_Audit_v0_27.zip and WRRA_Closed_Frame_and_Residual_Store_v0_28.zip.